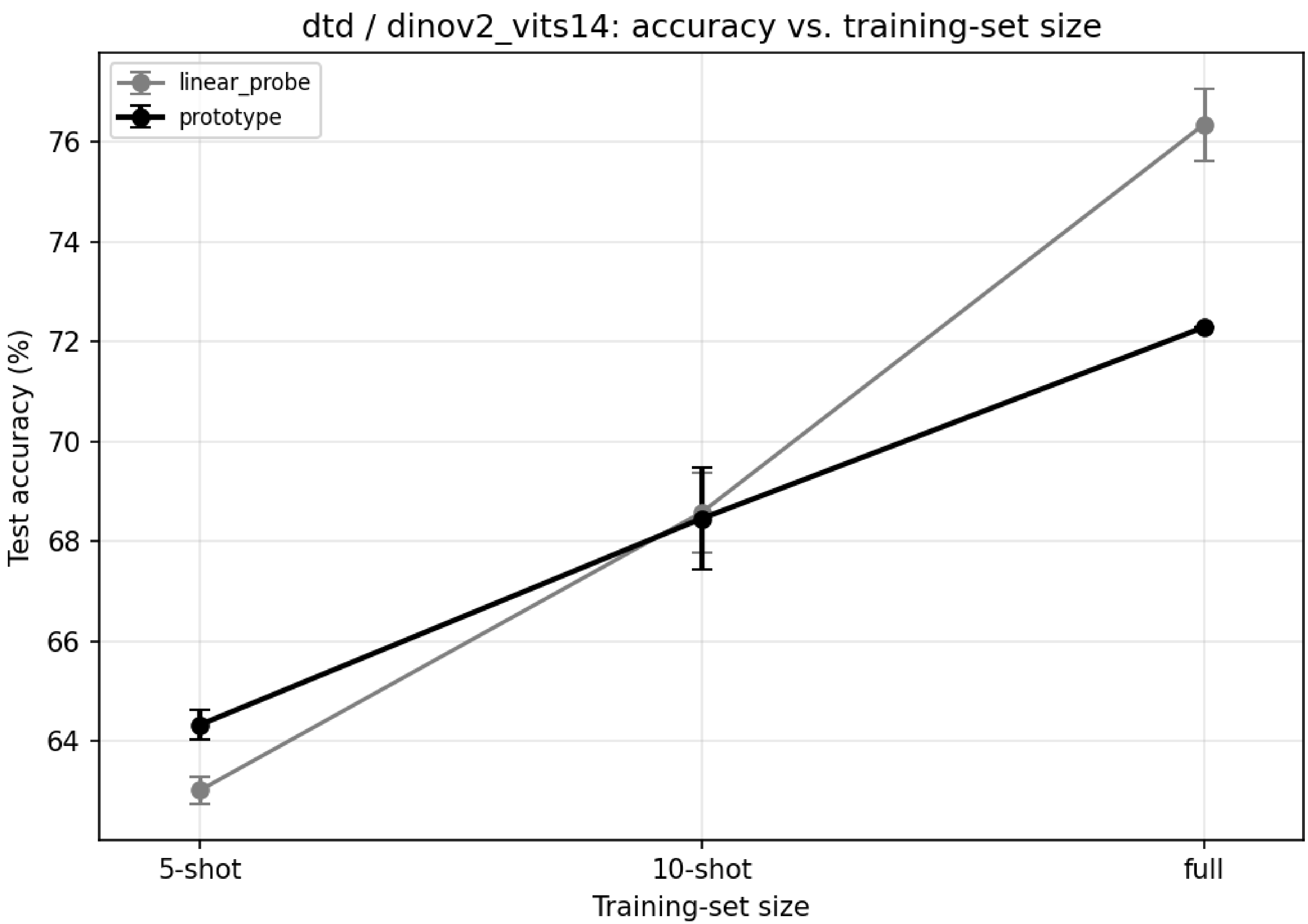
# Stage 1 and Stage 2 Results

| Dataset | Encoder | Method | T | K-shot | Runs | Test Accuracy | Delta |
|---|---|---|---|---|---|---|---|
| dtd | dinov2_vits14 | linear_probe | - | 5 | 3 | 63.01% +/- 0.27% | - |
| dtd | dinov2_vits14 | linear_probe | - | 10 | 3 | 68.58% +/- 0.80% | - |
| dtd | dinov2_vits14 | linear_probe | - | full | 3 | 76.35% +/- 0.72% | - |
| dtd | dinov2_vits14 | prototype | - | 5 | 3 | 64.33% +/- 0.30% | - |
| dtd | dinov2_vits14 | prototype | - | 10 | 3 | 68.46% +/- 1.01% | - |
| dtd | dinov2_vits14 | prototype | - | full | 1 | 72.29% | - |
| dtd | dinov2_vits14 | fm_standard | 4 | 5 | 3 | 61.31% +/- 0.75% | -3.01% +/- 0.46% |
| dtd | dinov2_vits14 | fm_standard | 12 | 5 | 3 | 61.70% +/- 0.83% | -2.62% +/- 0.55% |
| dtd | dinov2_vits14 | fm_standard | 4 | 10 | 3 | 65.60% +/- 0.59% | -2.85% +/- 1.45% |
| dtd | dinov2_vits14 | fm_standard | 12 | 10 | 3 | 65.92% +/- 0.57% | -2.54% +/- 1.48% |
| dtd | dinov2_vits14 | fm_standard | 4 | full | 1 | 74.63% | +2.34% |
| dtd | dinov2_vits14 | fm_standard | 12 | full | 1 | 74.79% | +2.50% |
| dtd | dinov2_vits14 | fm_rolled | 4 | 5 | 3 | 57.48% +/- 1.32% | -6.84% +/- 1.02% |
| dtd | dinov2_vits14 | fm_rolled | 12 | 5 | 3 | 57.50% +/- 1.04% | -6.83% +/- 0.73% |
| dtd | dinov2_vits14 | fm_rolled | 4 | 10 | 3 | 61.74% +/- 0.48% | -6.72% +/- 1.46% |
| dtd | dinov2_vits14 | fm_rolled | 12 | 10 | 3 | 61.74% +/- 0.27% | -6.72% +/- 1.28% |
| dtd | dinov2_vits14 | fm_rolled | 4 | full | 1 | 73.51% | +1.22% |
| dtd | dinov2_vits14 | fm_rolled | 12 | full | 1 | 72.93% | +0.64% |
| dtd | resnet18 | linear_probe | - | 5 | 3 | 45.80% +/- 0.75% | - |
| dtd | resnet18 | linear_probe | - | 10 | 3 | 52.39% +/- 0.74% | - |
| dtd | resnet18 | linear_probe | - | full | 3 | 62.55% +/- 0.51% | - |
| dtd | resnet18 | prototype | - | 5 | 3 | 46.93% +/- 0.32% | - |
| dtd | resnet18 | prototype | - | 10 | 3 | 51.97% +/- 0.54% | - |
| dtd | resnet18 | prototype | - | full | 1 | 58.78% | - |
| dtd | resnet18 | fm_standard | 4 | 5 | 3 | 44.82% +/- 1.45% | -2.11% +/- 1.19% |
| dtd | resnet18 | fm_standard | 12 | 5 | 3 | 45.14% +/- 1.36% | -1.79% +/- 1.07% |

| Dataset | Encoder | Method | T | K-shot | Runs | Test Accuracy | Delta |
|---|---|---|---|---|---|---|---|
| dtd | resnet18 | fm_standard | 4 | 10 | 3 | 48.65% +/- 1.02% | -3.32% +/- 0.52% |
| dtd | resnet18 | fm_standard | 12 | 10 | 3 | 49.13% +/- 1.17% | -2.84% +/- 0.65% |
| dtd | resnet18 | fm_standard | 4 | full | 1 | 59.52% | +0.74% |
| dtd | resnet18 | fm_standard | 12 | full | 1 | 59.57% | +0.80% |
| dtd | resnet18 | fm_rolled | 4 | 5 | 3 | 40.51% +/- 0.75% | -6.42% +/- 0.45% |
| dtd | resnet18 | fm_rolled | 12 | 5 | 3 | 40.60% +/- 0.85% | -6.33% +/- 0.53% |
| dtd | resnet18 | fm_rolled | 4 | 10 | 3 | 40.62% +/- 1.00% | -11.35% +/- 1.20% |
| dtd | resnet18 | fm_rolled | 12 | 10 | 3 | 41.65% +/- 0.90% | -10.32% +/- 0.88% |
| dtd | resnet18 | fm_rolled | 4 | full | 1 | 56.06% | -2.71% |
| dtd | resnet18 | fm_rolled | 12 | full | 1 | 56.01% | -2.77% |
| flowers102 | resnet18 | linear_probe | - | 5 | 3 | 75.13% +/- 0.22% | - |
| flowers102 | resnet18 | linear_probe | - | 10 | 3 | 83.22% +/- 0.14% | - |
| flowers102 | resnet18 | linear_probe | - | full | 3 | 83.22% +/- 0.14% | - |
| flowers102 | resnet18 | prototype | - | 5 | 3 | 69.62% +/- 0.32% | - |
| flowers102 | resnet18 | prototype | - | 10 | 3 | 75.22% +/- 0.00% | - |
| flowers102 | resnet18 | prototype | - | full | 1 | 75.22% | - |
| flowers102 | resnet18 | fm_standard | 4 | 5 | 3 | 69.19% +/- 1.02% | -0.43% +/- 0.85% |
| flowers102 | resnet18 | fm_standard | 12 | 5 | 3 | 69.59% +/- 1.01% | -0.02% +/- 0.82% |
| flowers102 | resnet18 | fm_standard | 4 | 10 | 3 | 76.83% +/- 0.11% | +1.62% +/- 0.11% |
| flowers102 | resnet18 | fm_standard | 12 | 10 | 3 | 77.17% +/- 0.21% | +1.95% +/- 0.21% |
| flowers102 | resnet18 | fm_standard | 4 | full | 1 | 76.79% | +1.58% |
| flowers102 | resnet18 | fm_standard | 12 | full | 1 | 76.94% | +1.72% |
| flowers102 | resnet18 | fm_rolled | 4 | 5 | 3 | 62.54% +/- 1.32% | -7.07% +/- 1.12% |
| flowers102 | resnet18 | fm_rolled | 12 | 5 | 3 | 63.07% +/- 0.68% | -6.54% +/- 0.40% |
| flowers102 | resnet18 | fm_rolled | 4 | 10 | 3 | 72.46% +/- 0.38% | -2.76% +/- 0.38% |
| flowers102 | resnet18 | fm_rolled | 12 | 10 | 3 | 72.65% +/- 0.37% | -2.57% +/- 0.37% |
| flowers102 | resnet18 | fm_rolled | 4 | full | 1 | 72.89% | -2.33% |
| flowers102 | resnet18 | fm_rolled | 12 | full | 1 | 73.07% | -2.15% |

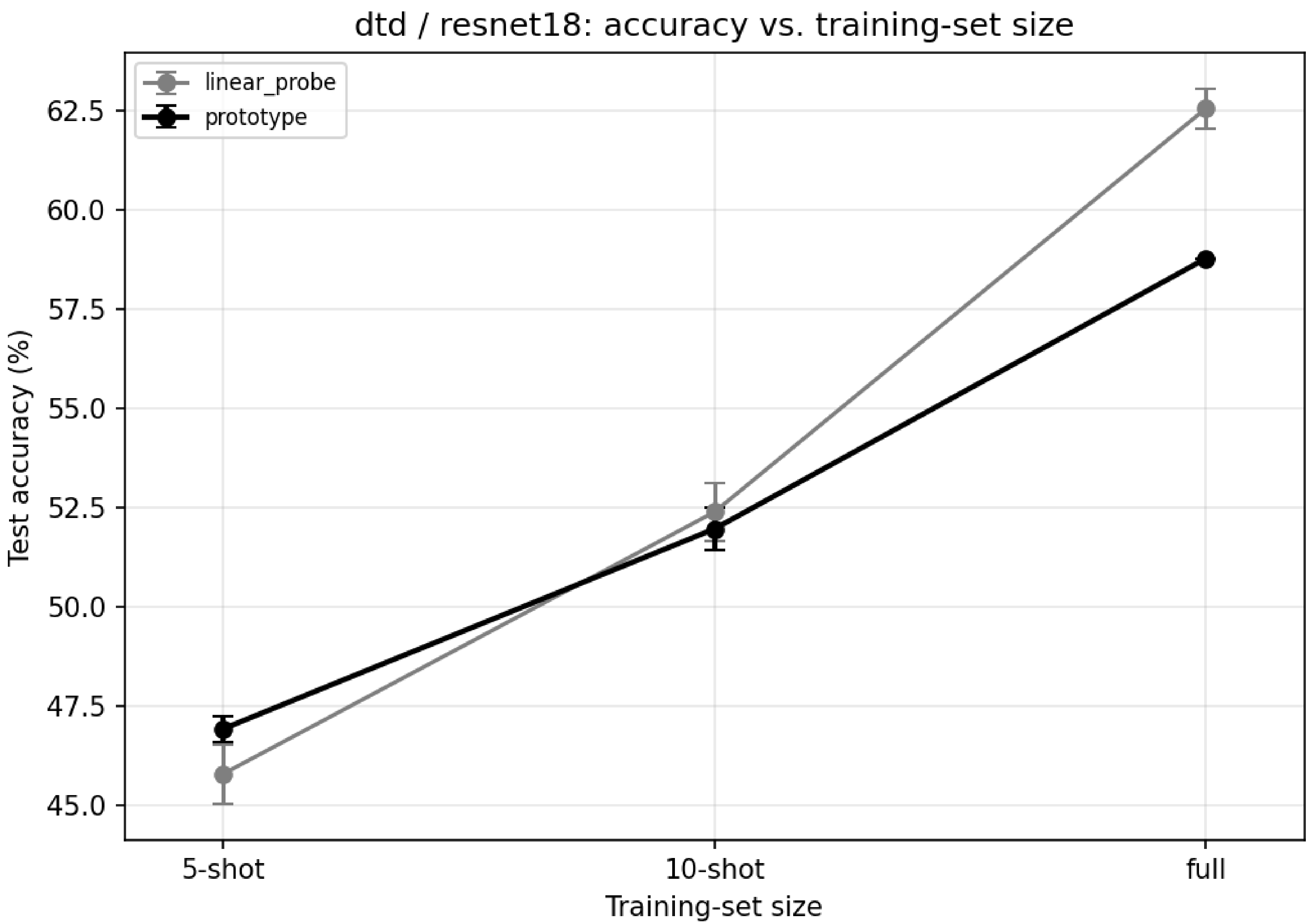
**Accuracy vs. training-set size**

***dtd / dinov2_vits14***



## Accuracy vs. training-set size

***dtd / resnet18***

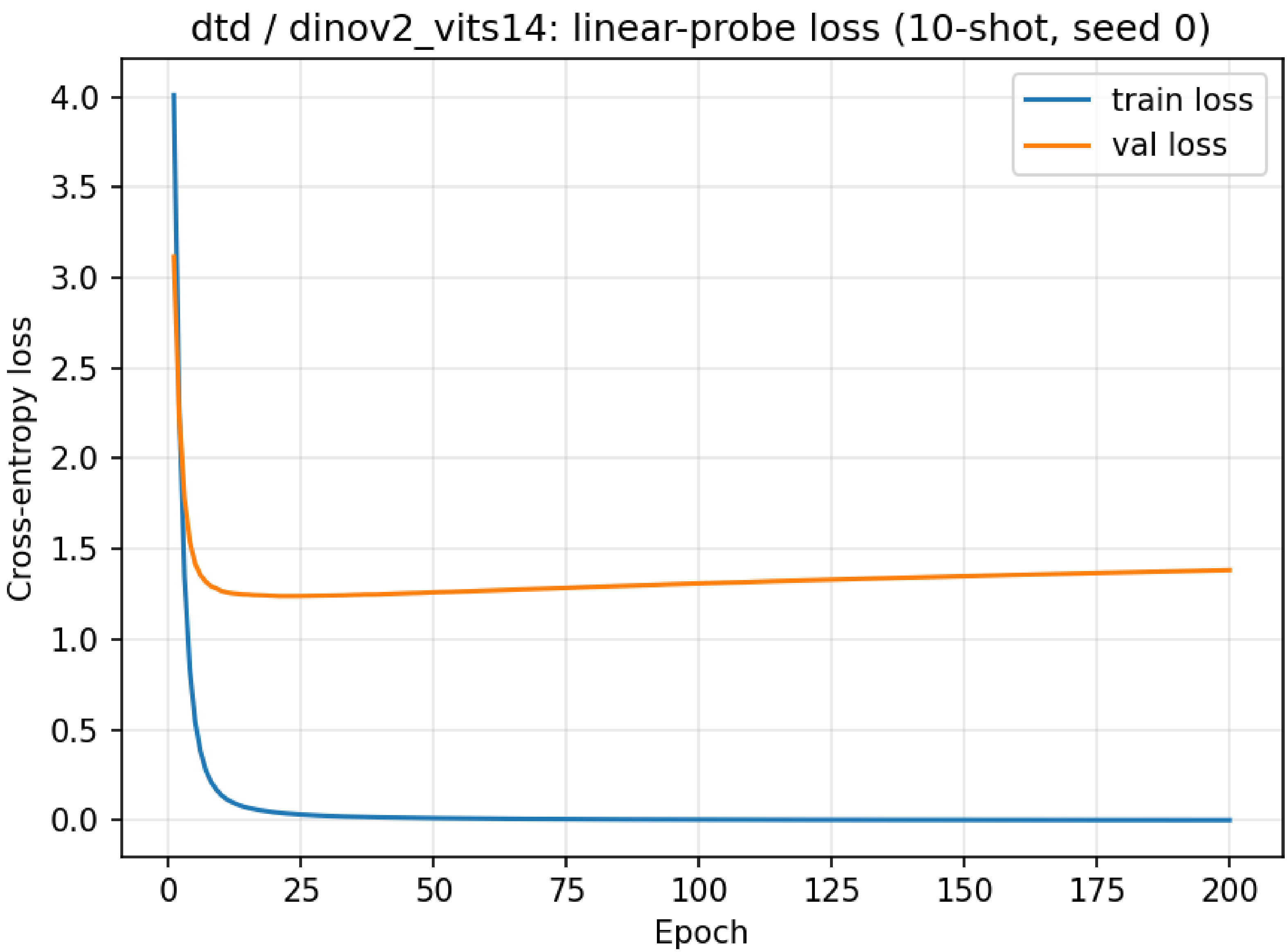


## Accuracy vs. training-set size

***flowers102 / resnet18***

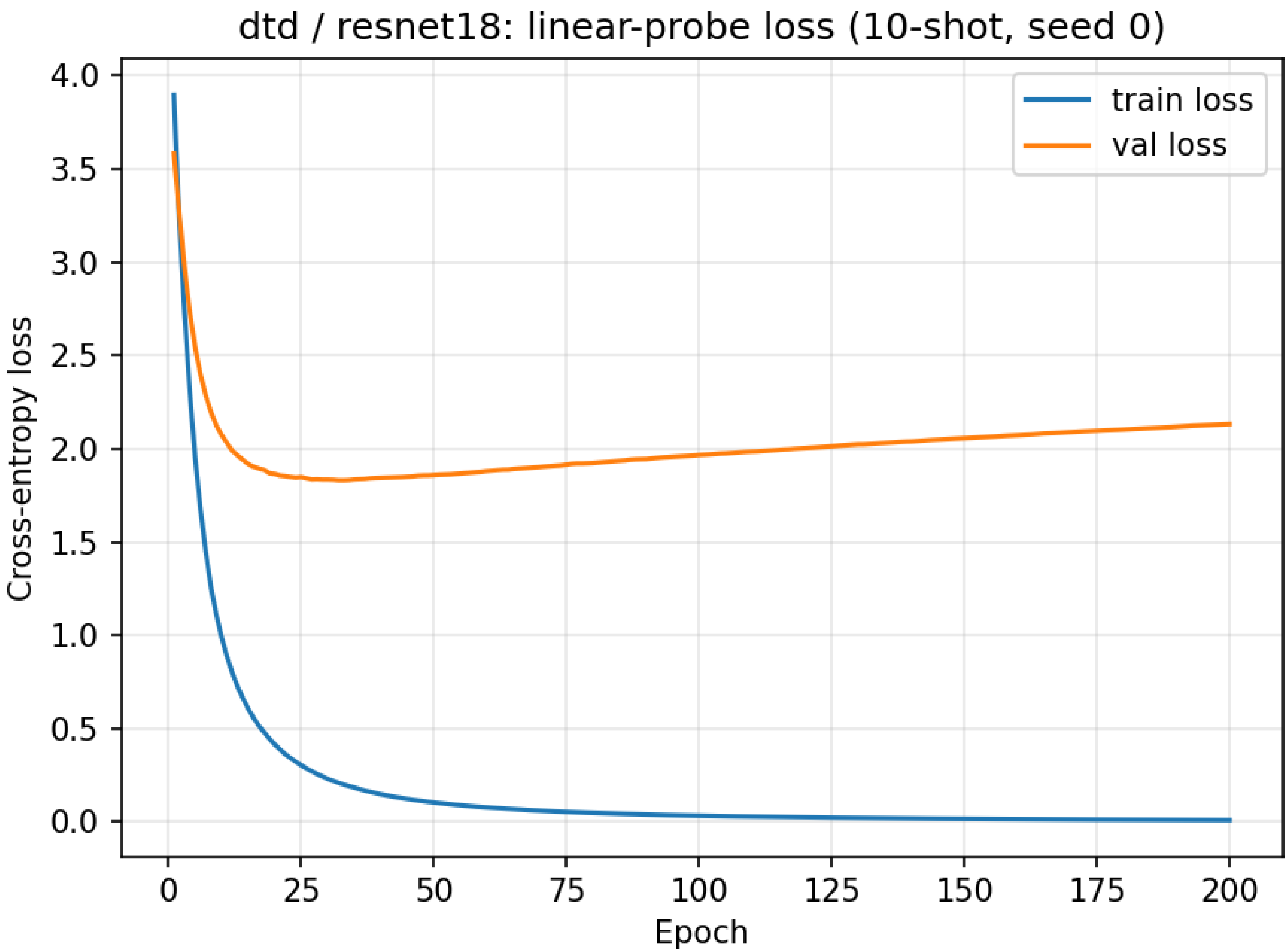
**Training / validation loss (10-shot, seed 0)**

***dtd / dinov2_vits14***



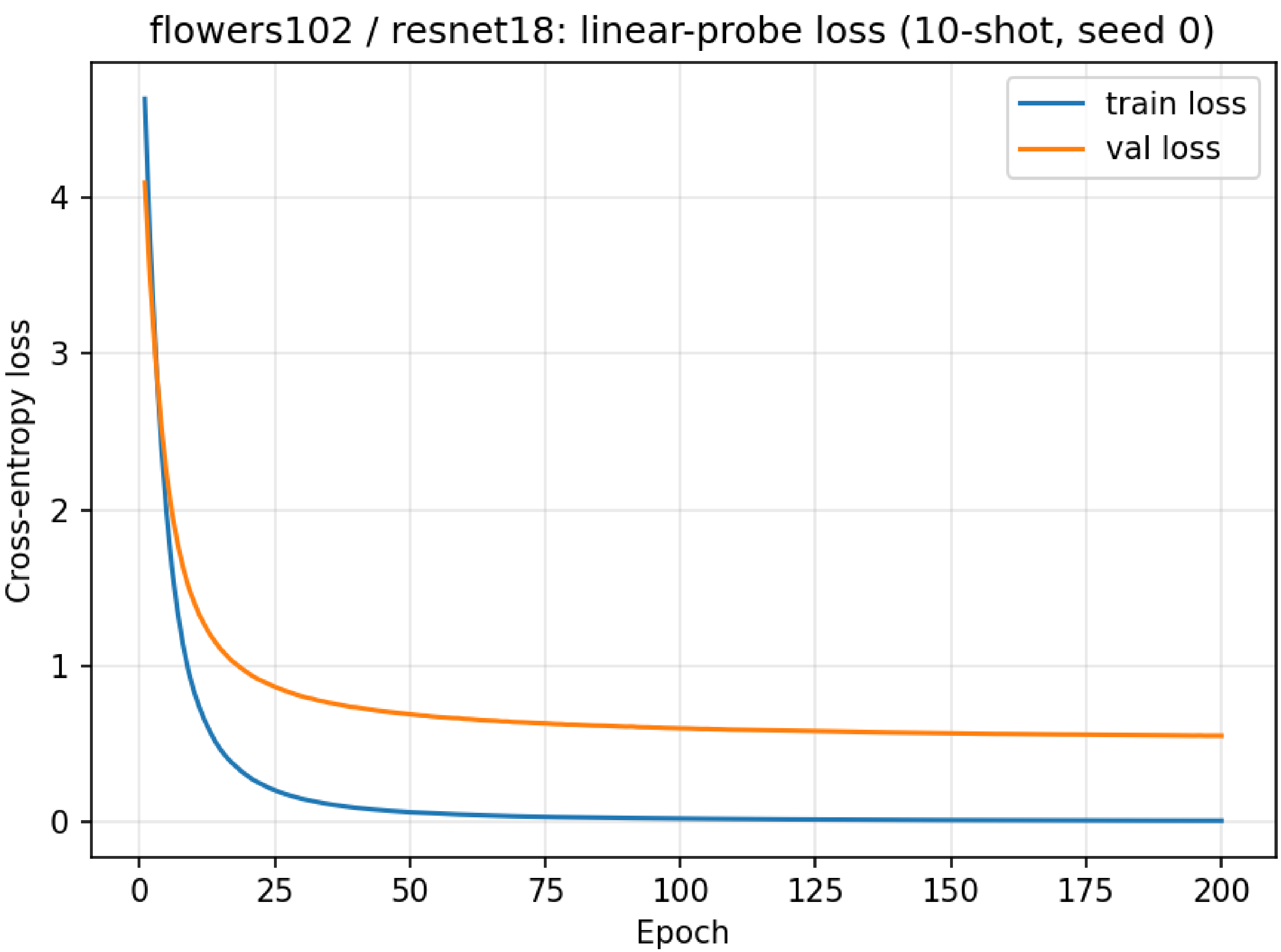
**Training / validation loss (10-shot, seed 0)**

***dtd / resnet18***



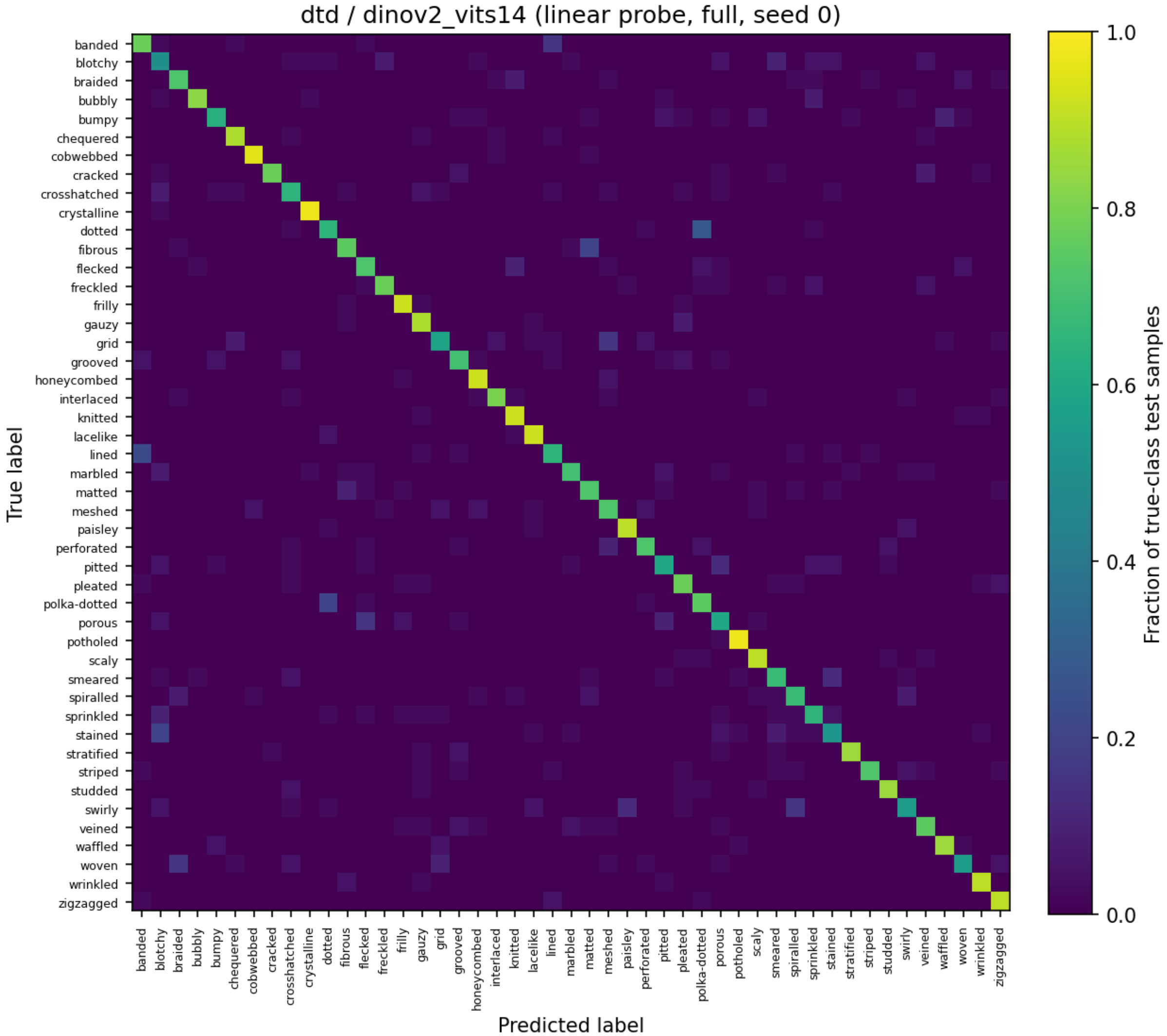
**Training / validation loss (10-shot, seed 0)**

***flowers102 / resnet18***



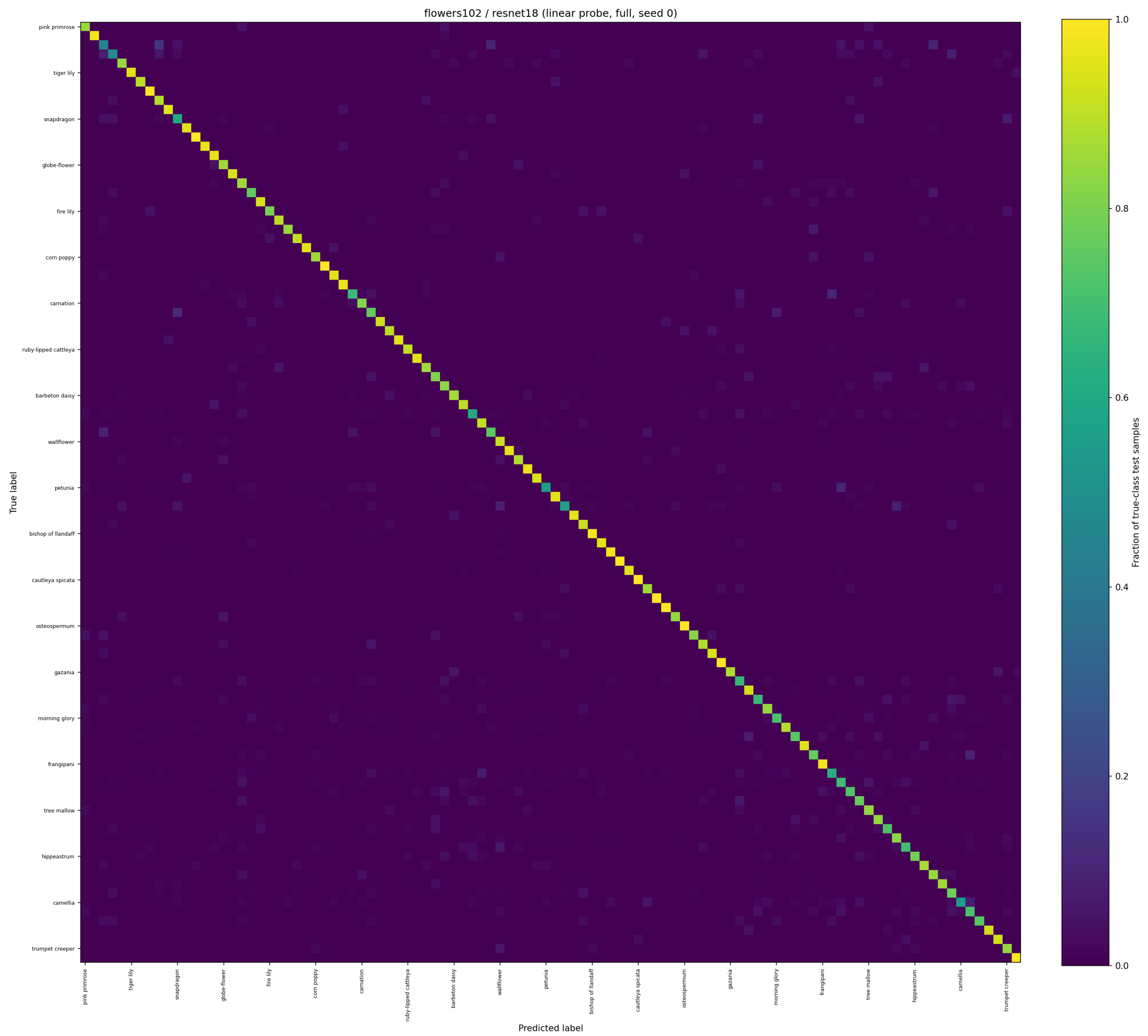
# Confusion matrices (row-normalized)

### *dtd / dinov2_vits14*



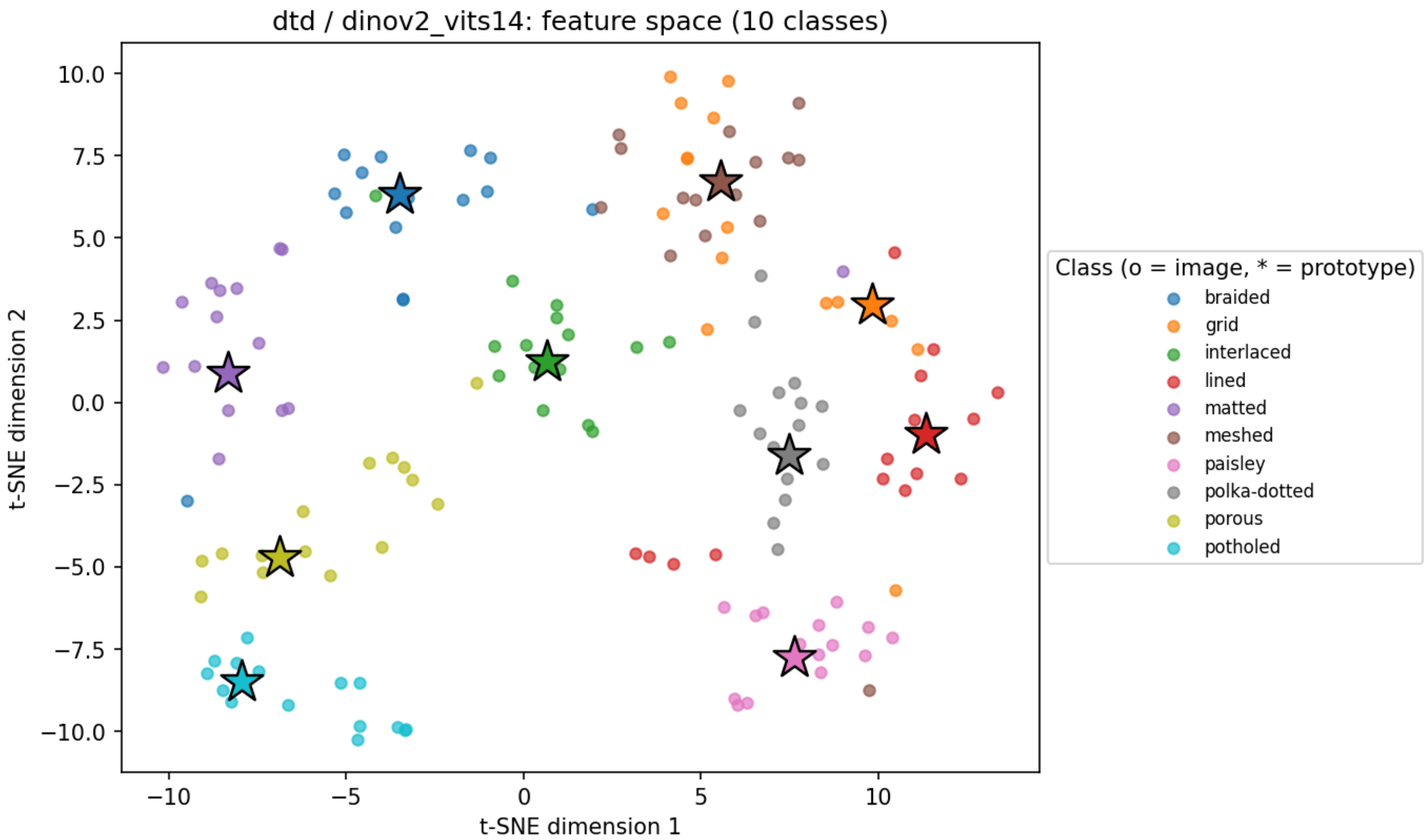
# Confusion matrices (row-normalized)

## *flowers102 / resnet18*



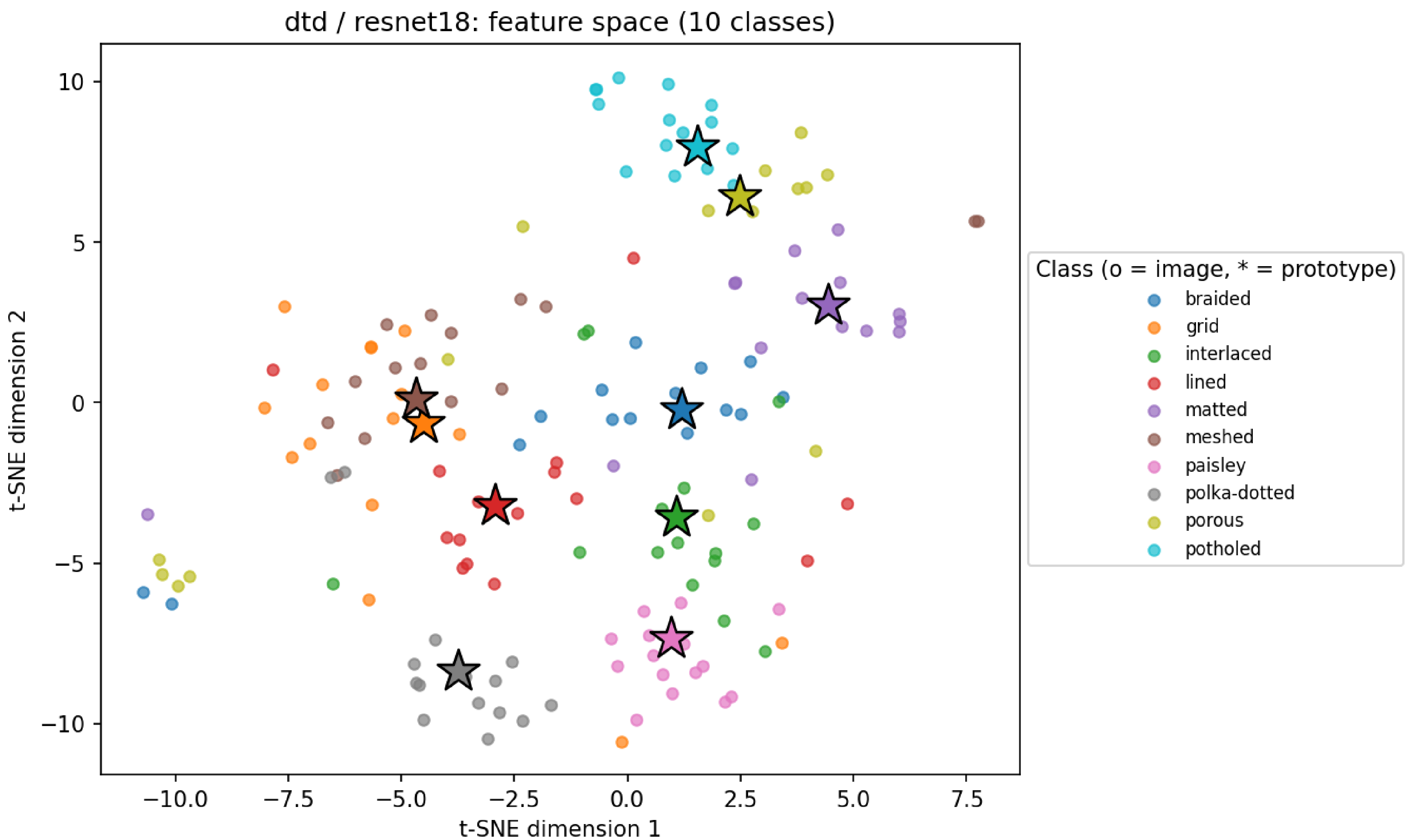
## Feature-space visualizations (t-SNE)

### *dtd / dinov2_vits14*



## Feature-space visualizations (t-SNE)

### *dtd / resnet18*

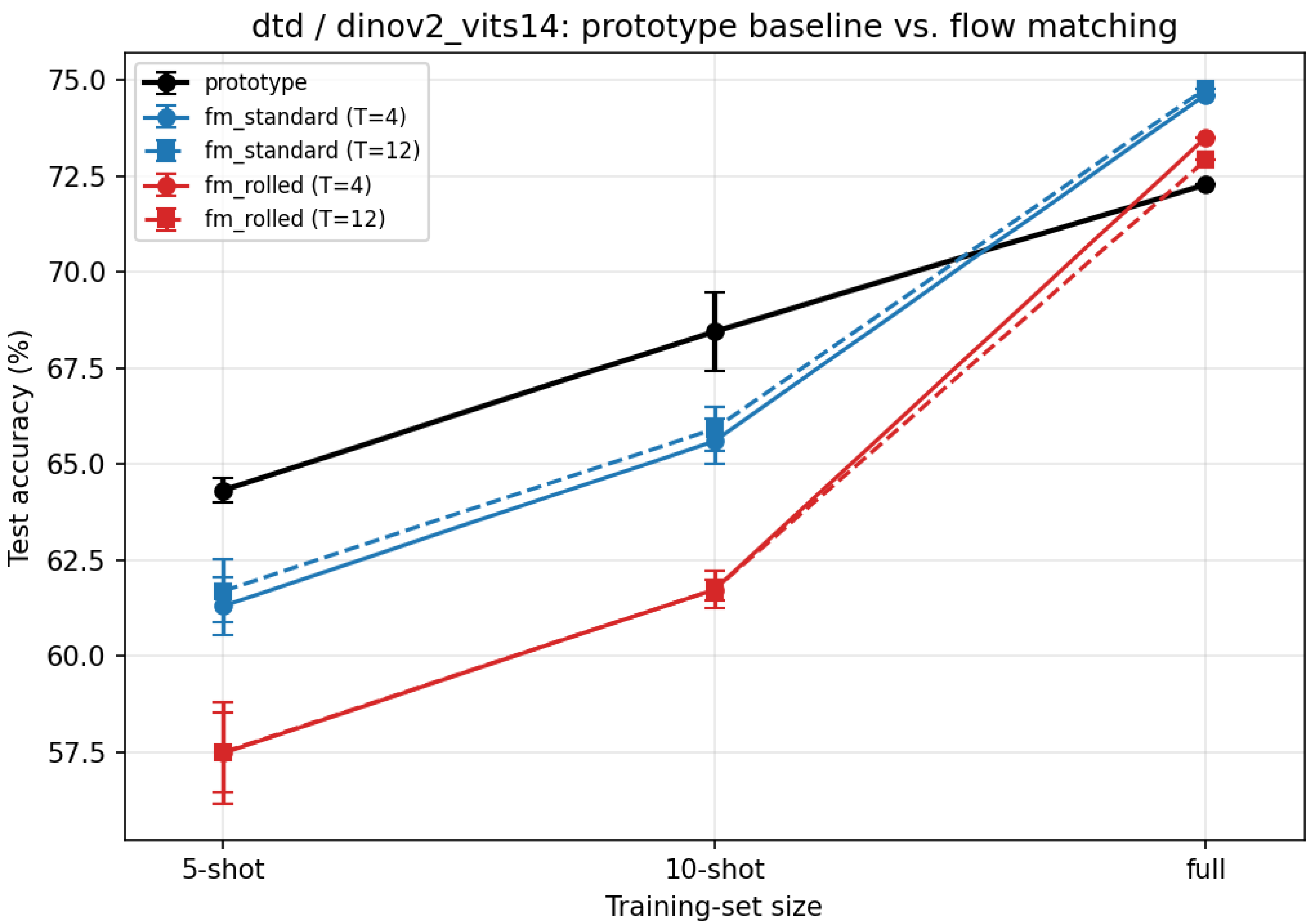


## Feature-space visualizations (t-SNE)

***flowers102 / resnet18***

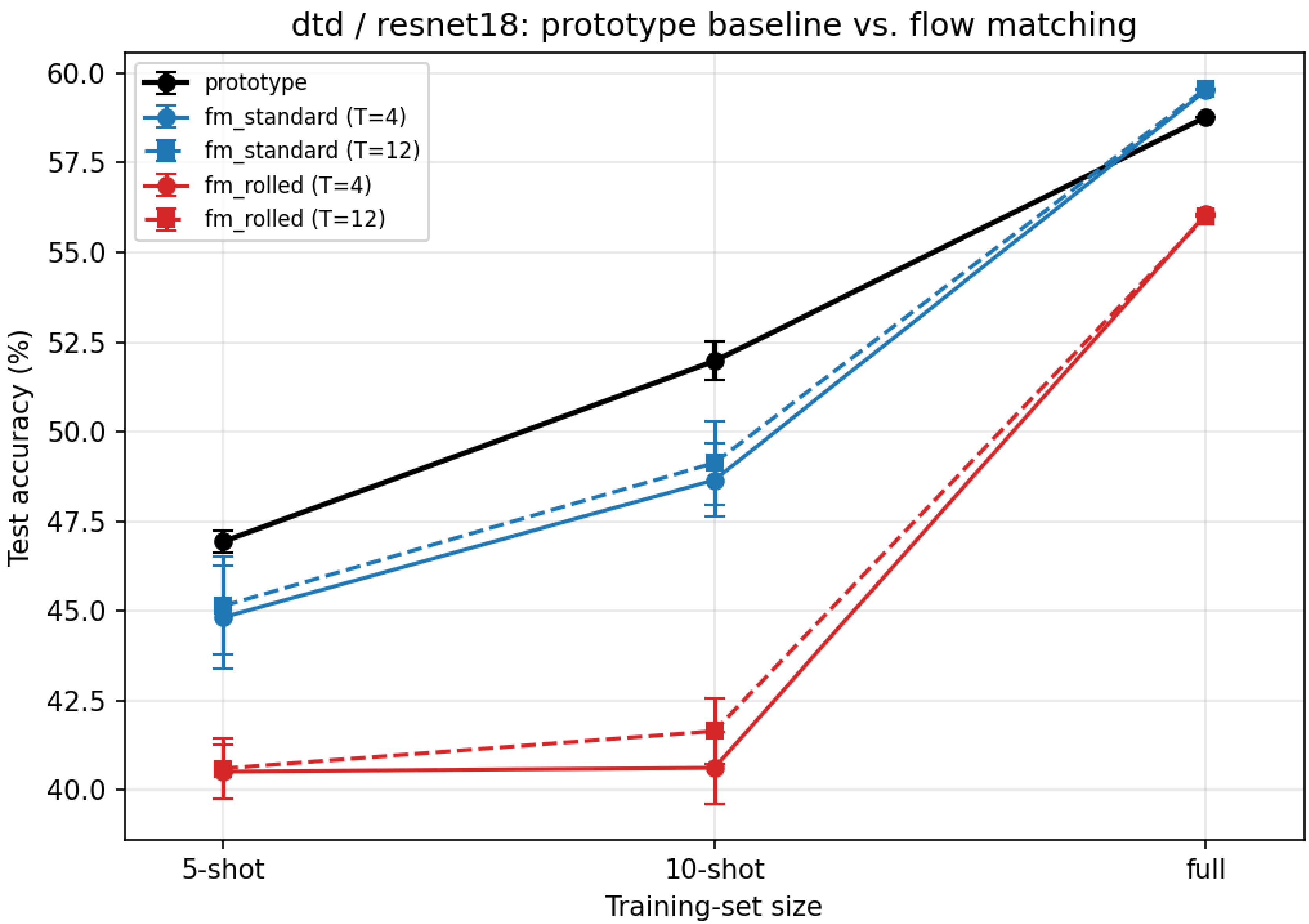
**Stage 2: accuracy vs. training-set size**

***dtd / dinov2_vits14***



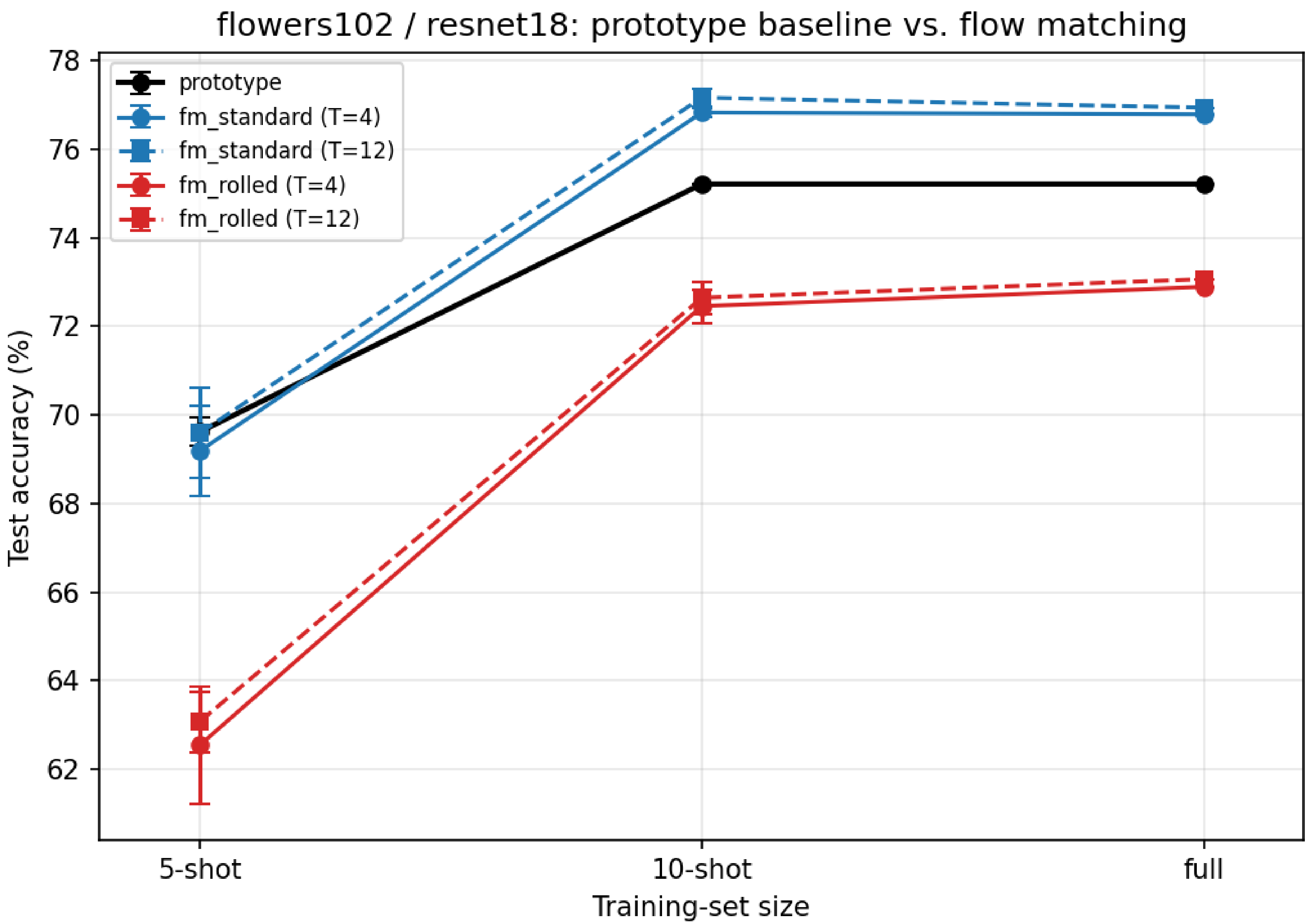
**Stage 2: accuracy vs. training-set size**

***dtd / resnet18***



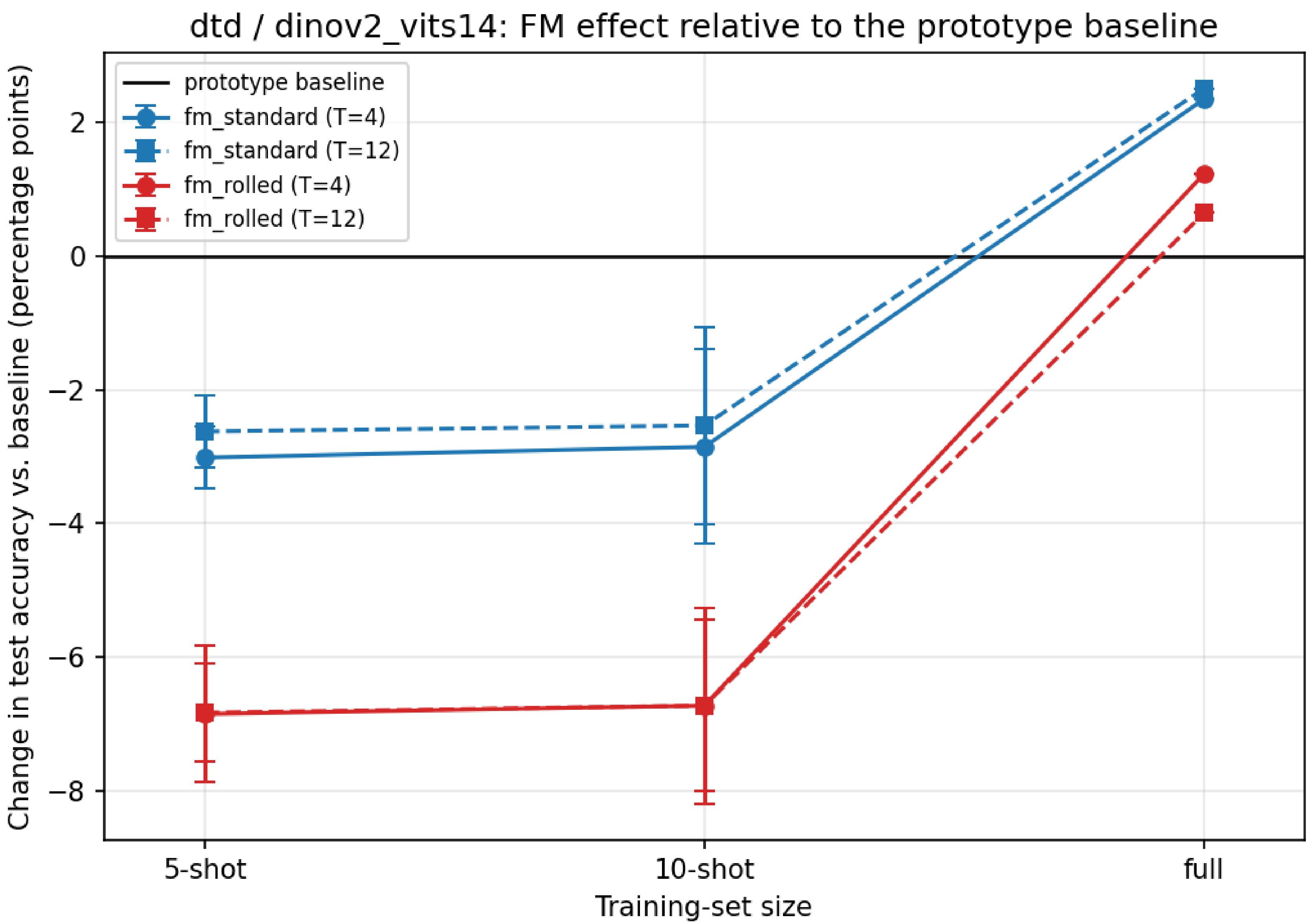
**Stage 2: accuracy vs. training-set size**

***flowers102 / resnet18***



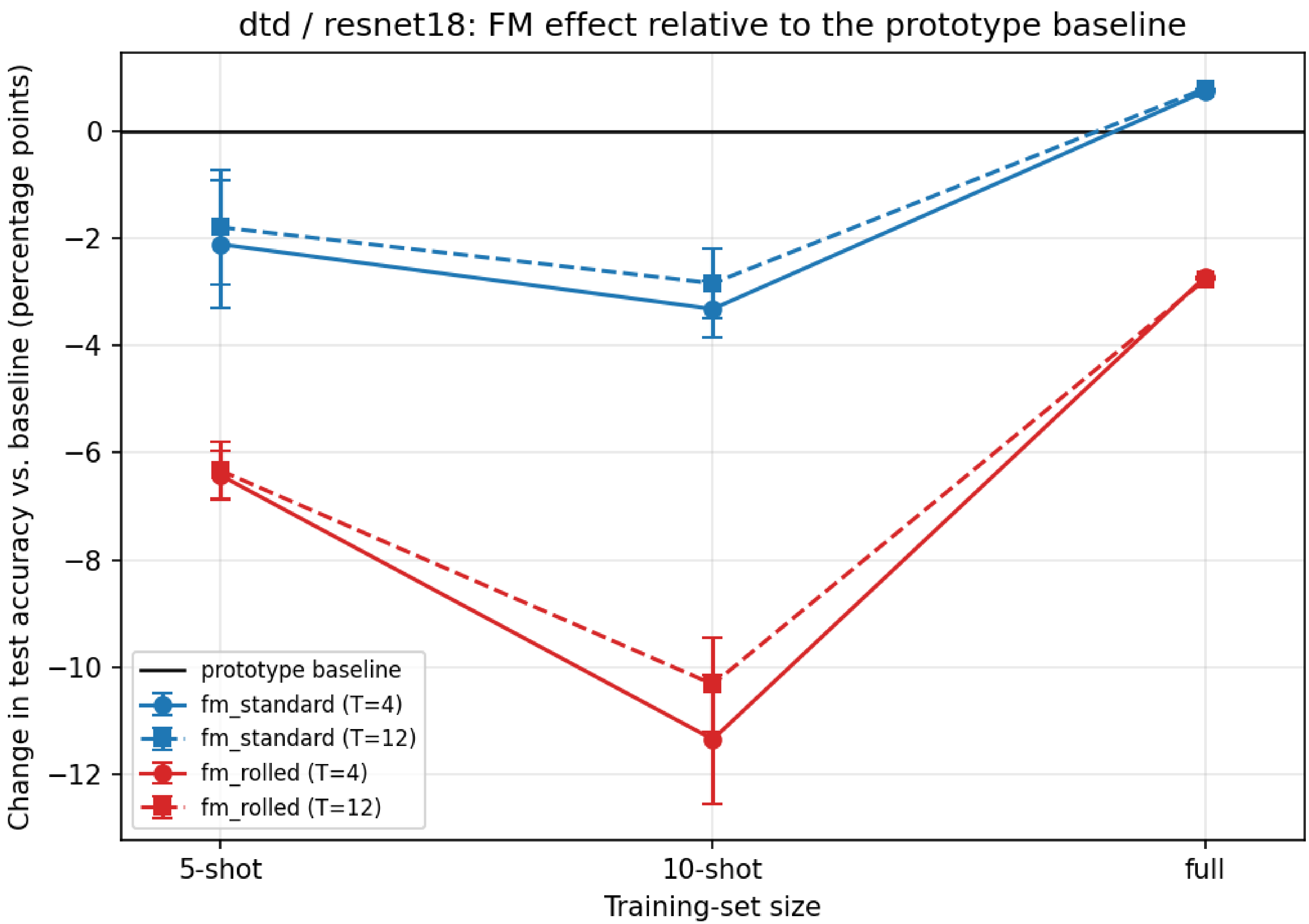
**Stage 2: change in accuracy relative to the prototype baseline**

***dtd / dinov2_vits14***



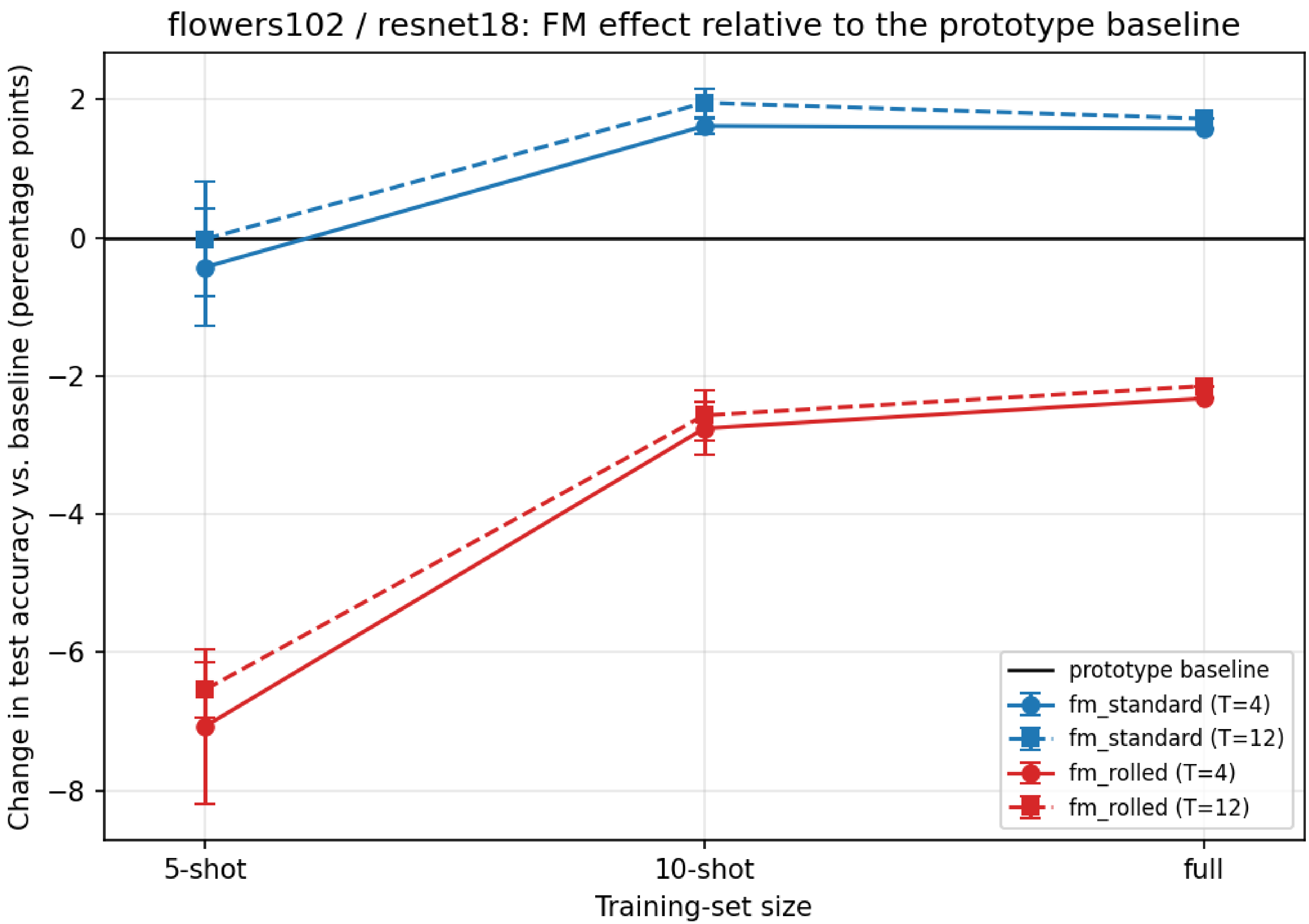
**Stage 2: change in accuracy relative to the prototype baseline**

***dtd / resnet18***



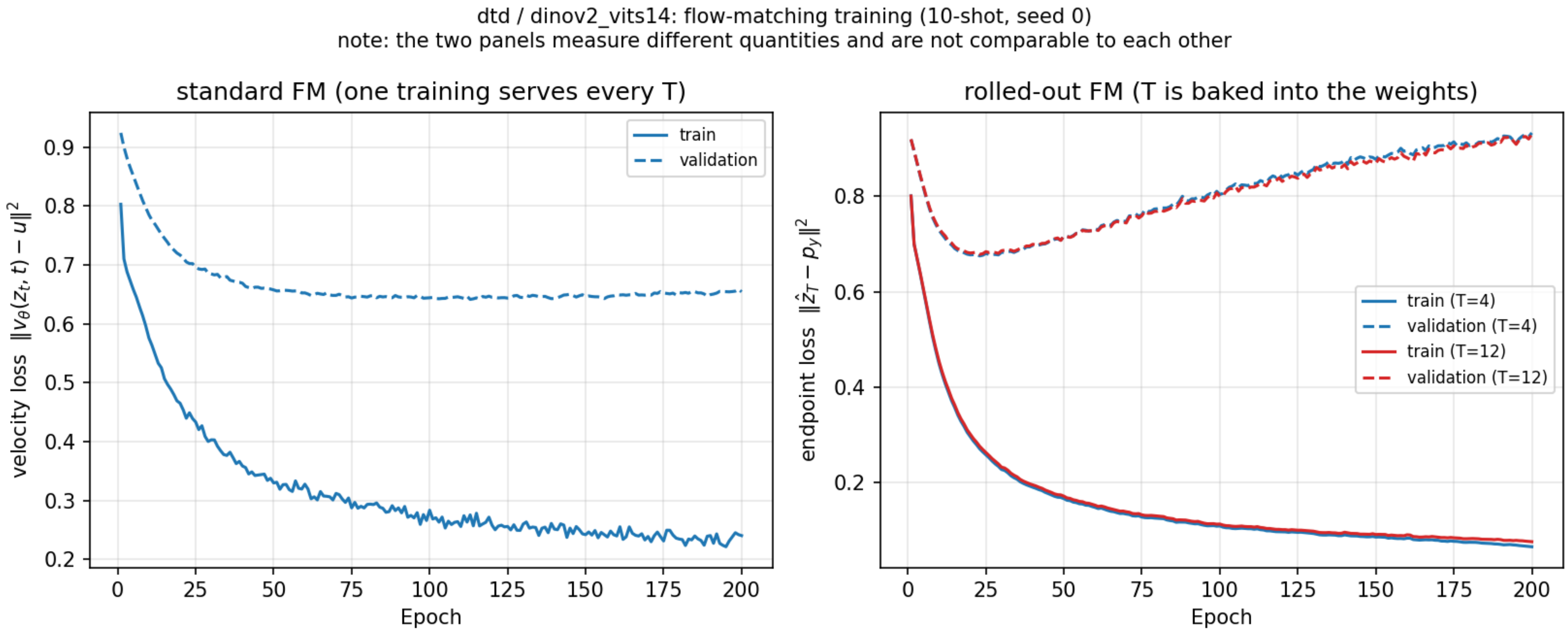
**Stage 2: change in accuracy relative to the prototype baseline**

***flowers102 / resnet18***



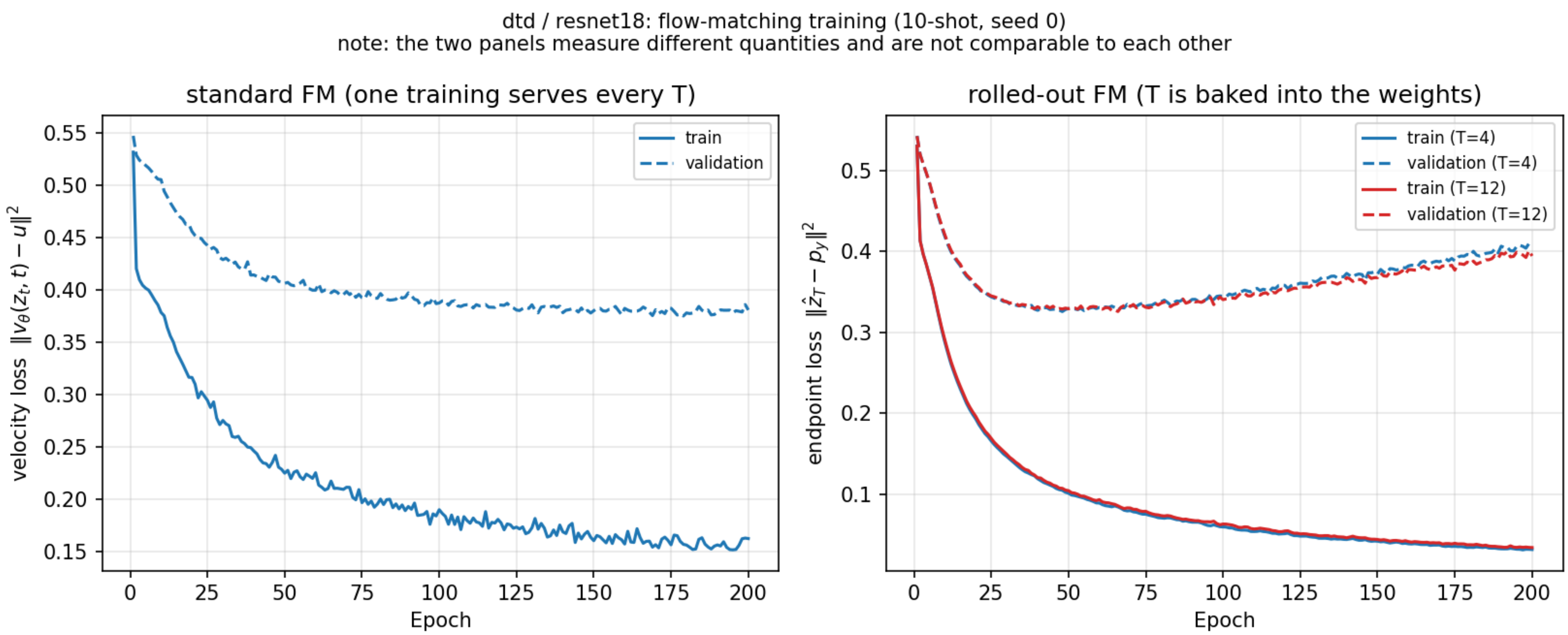
## Stage 2: flow-matching training curves (10-shot, seed 0)

### *dtd / dinov2_vits14*



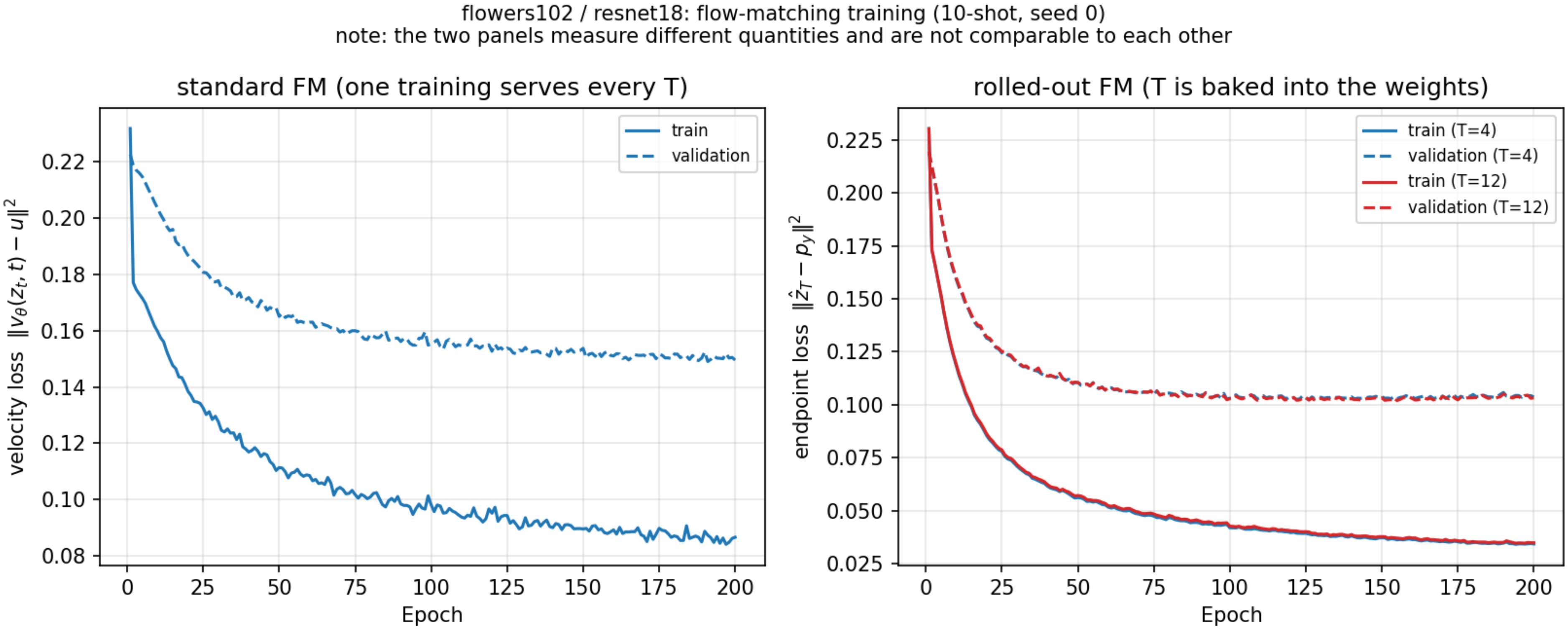
## Stage 2: flow-matching training curves (10-shot, seed 0)

### *dtd / resnet18*



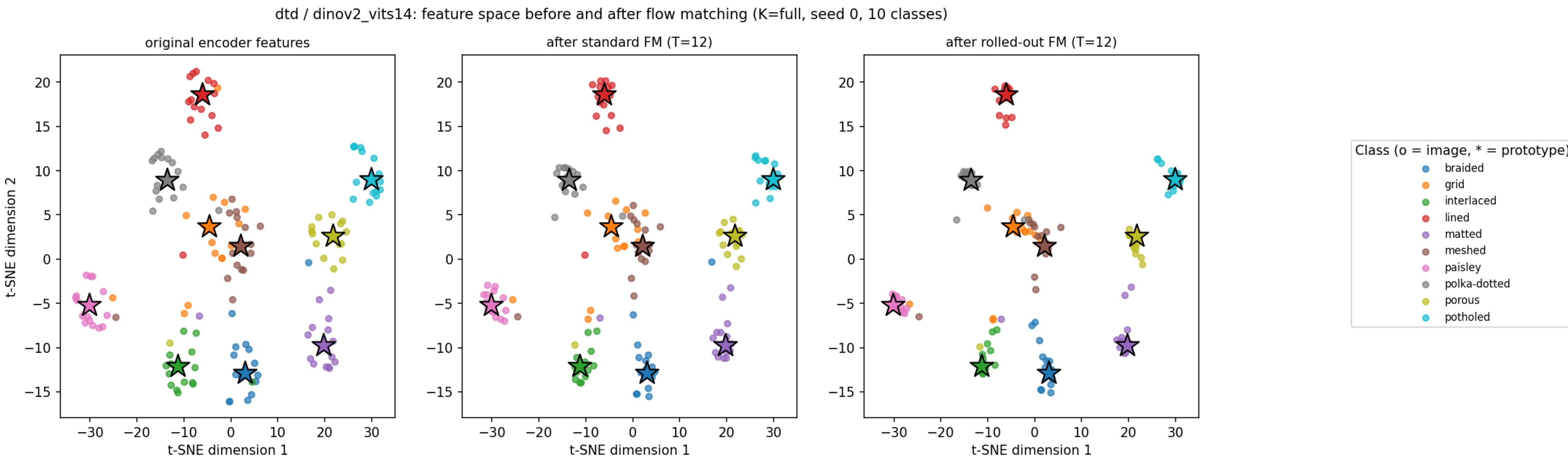
## Stage 2: flow-matching training curves (10-shot, seed 0)

### *flowers102 / resnet18*



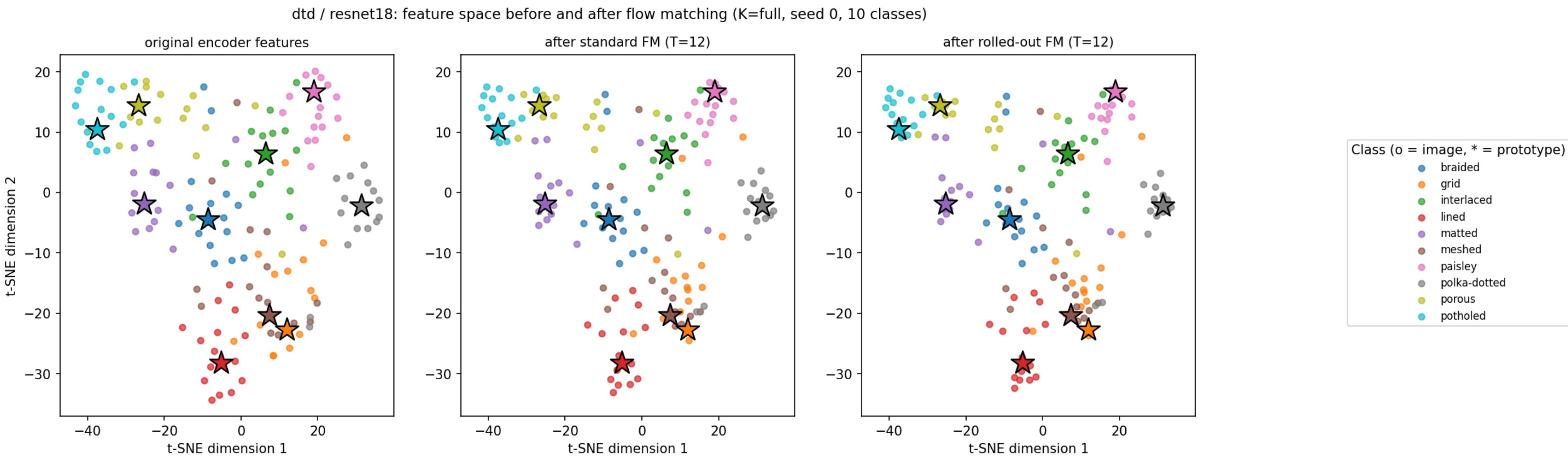
# Stage 2: feature space before and after flow matching (t-SNE)

## *dtd / dinov2_vits14*



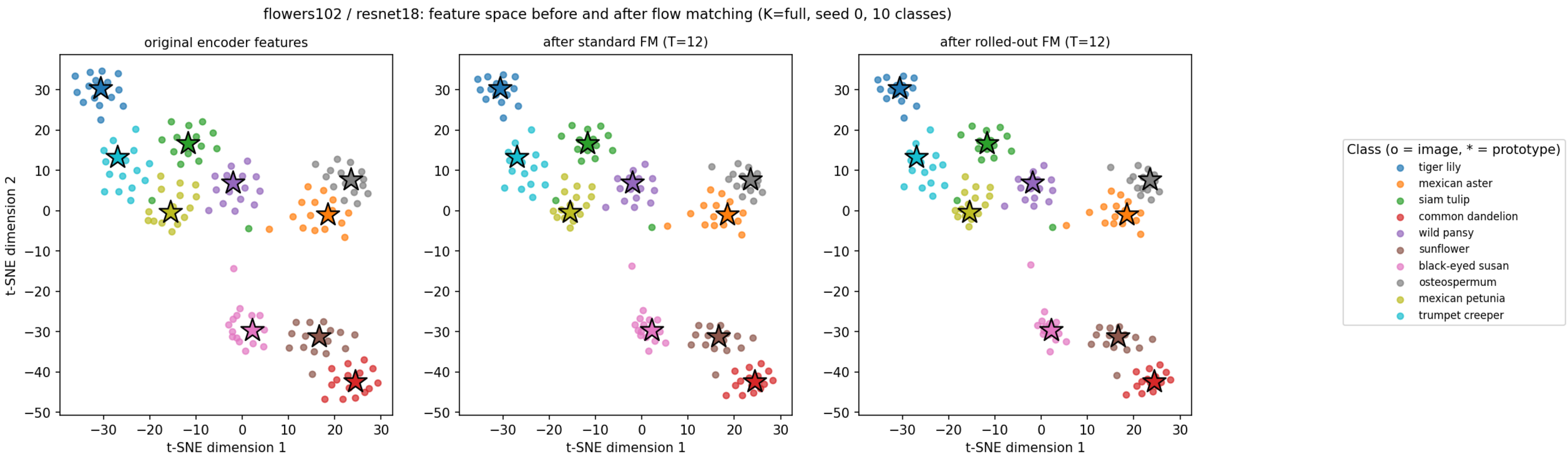
# Stage 2: feature space before and after flow matching (t-SNE)

## *dtd / resnet18*



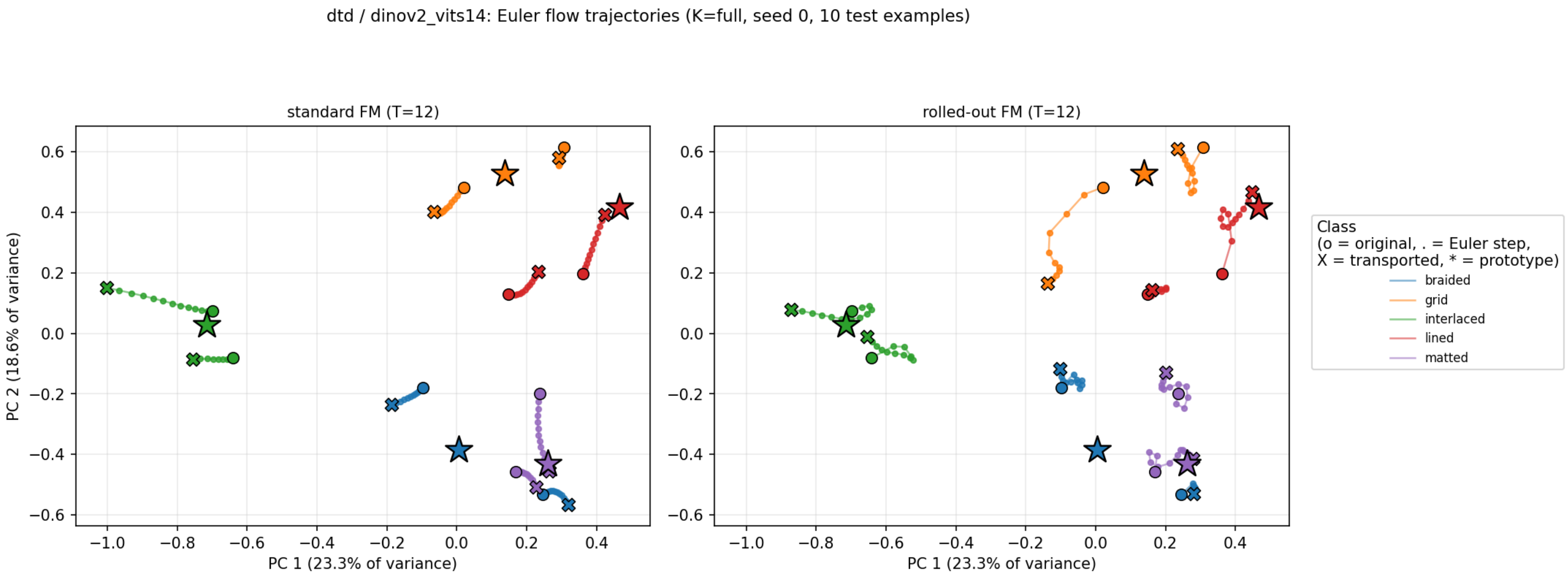
# Stage 2: feature space before and after flow matching (t-SNE)

## *flowers102 / resnet18*



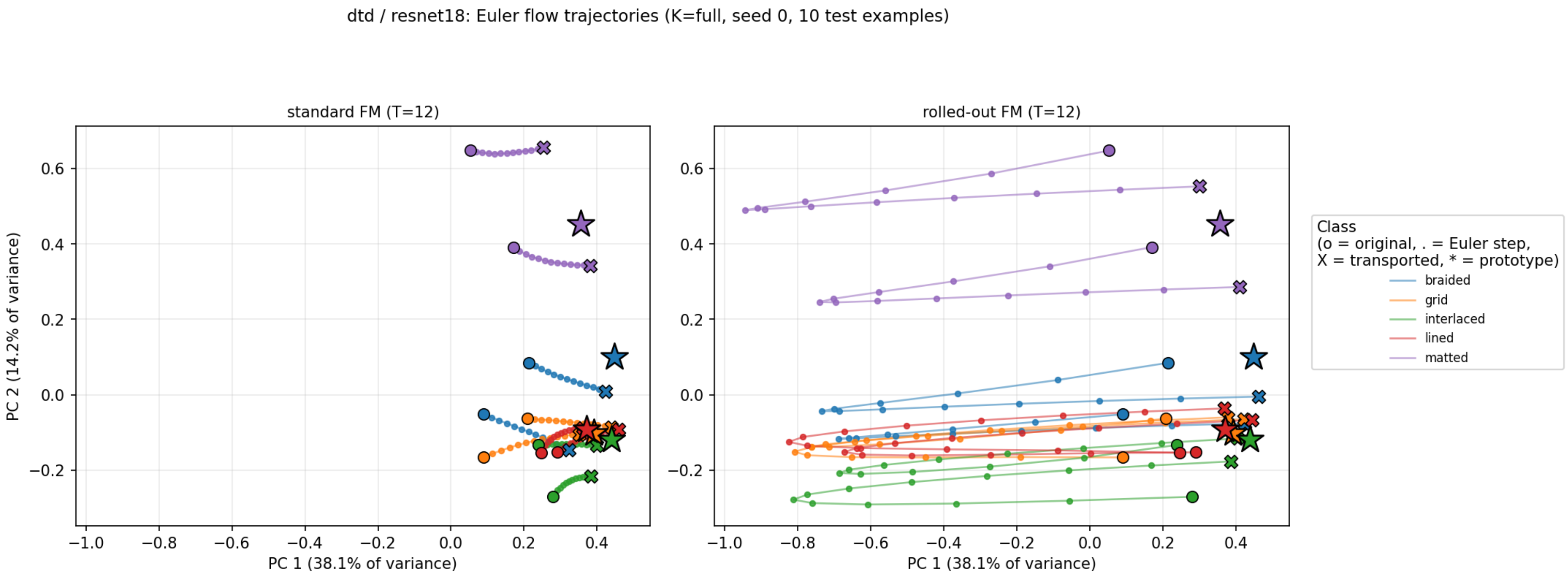
## Stage 2: flow trajectories (PCA)

### *dtd / dinov2_vits14*



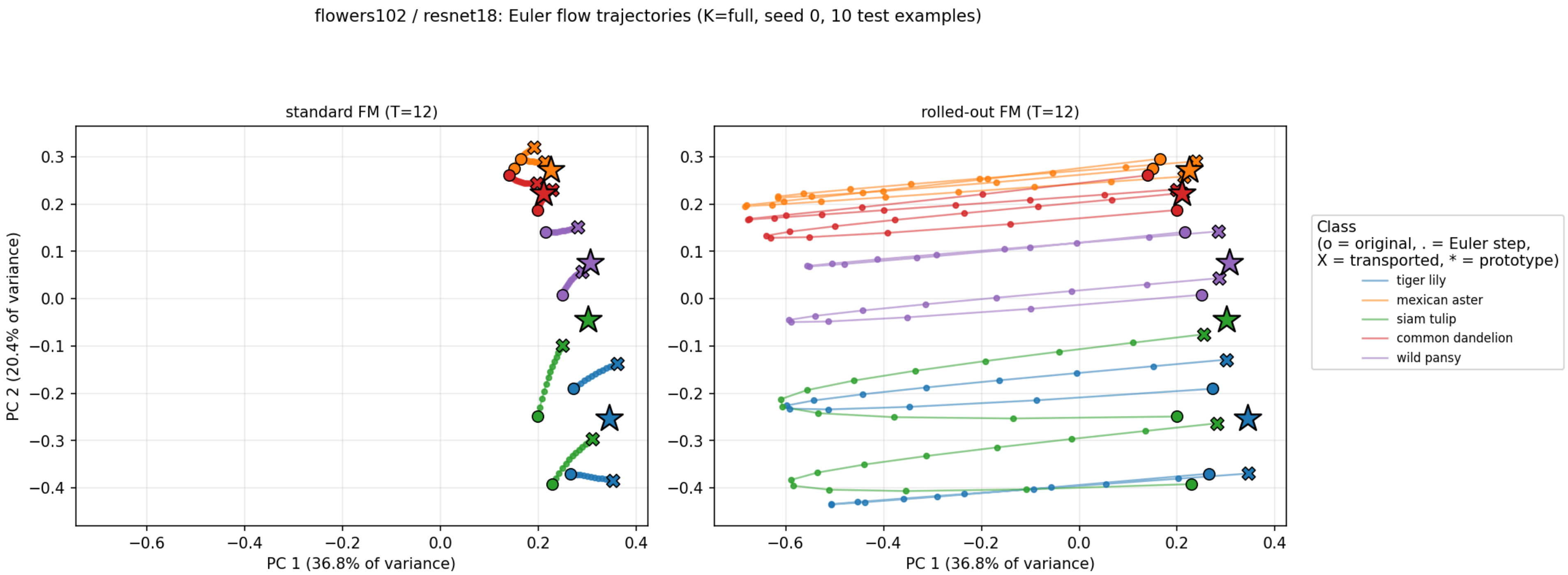
# Stage 2: flow trajectories (PCA)

## *dtd / resnet18*



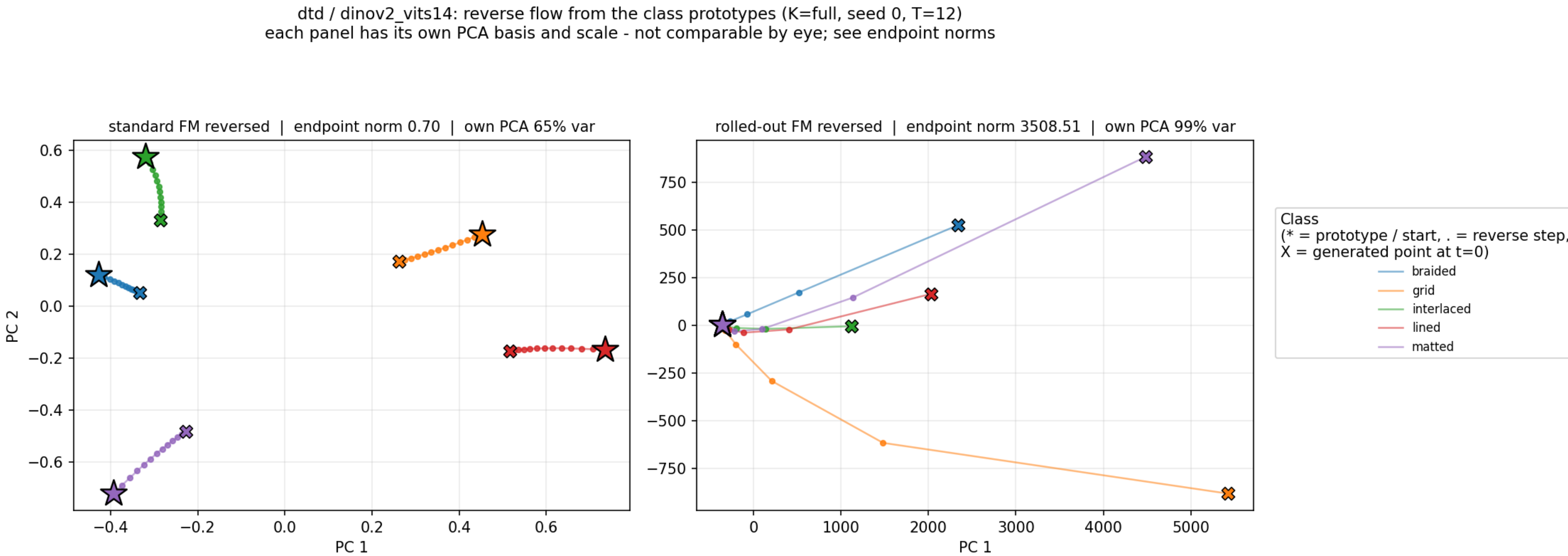
# Stage 2: flow trajectories (PCA)

## *flowers102 / resnet18*



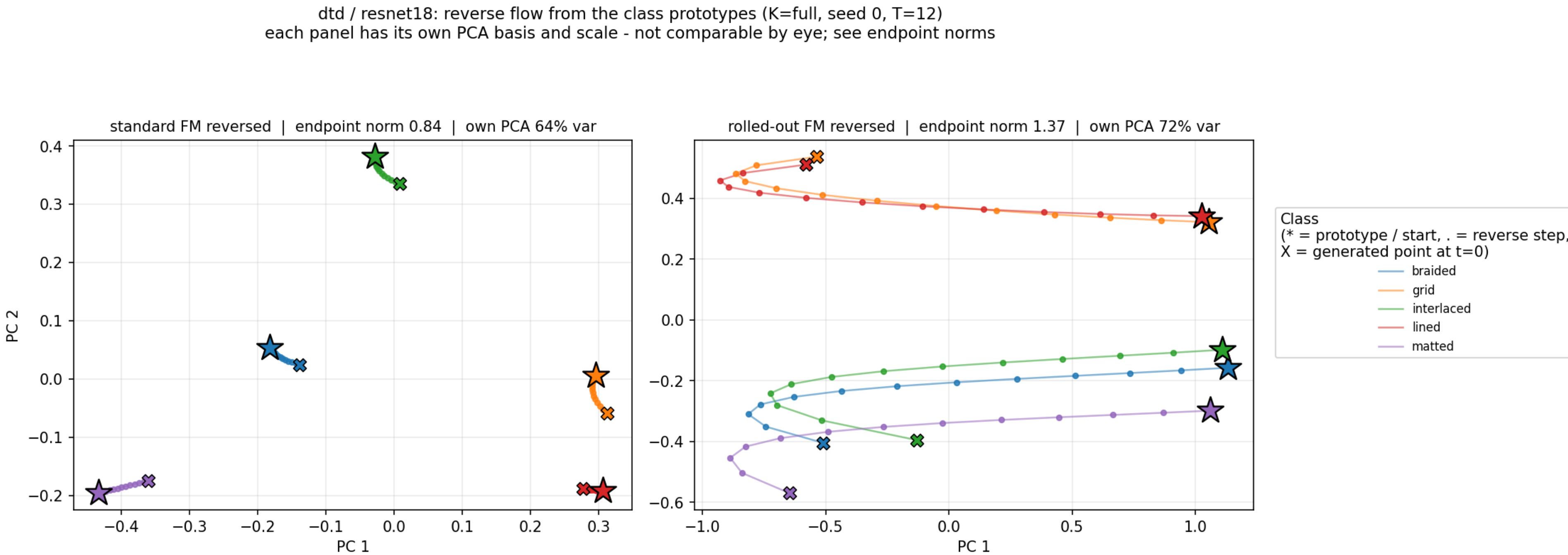
## Stage 2: reverse flow from the class prototypes (PCA)

### *dtd / dinov2_vits14*



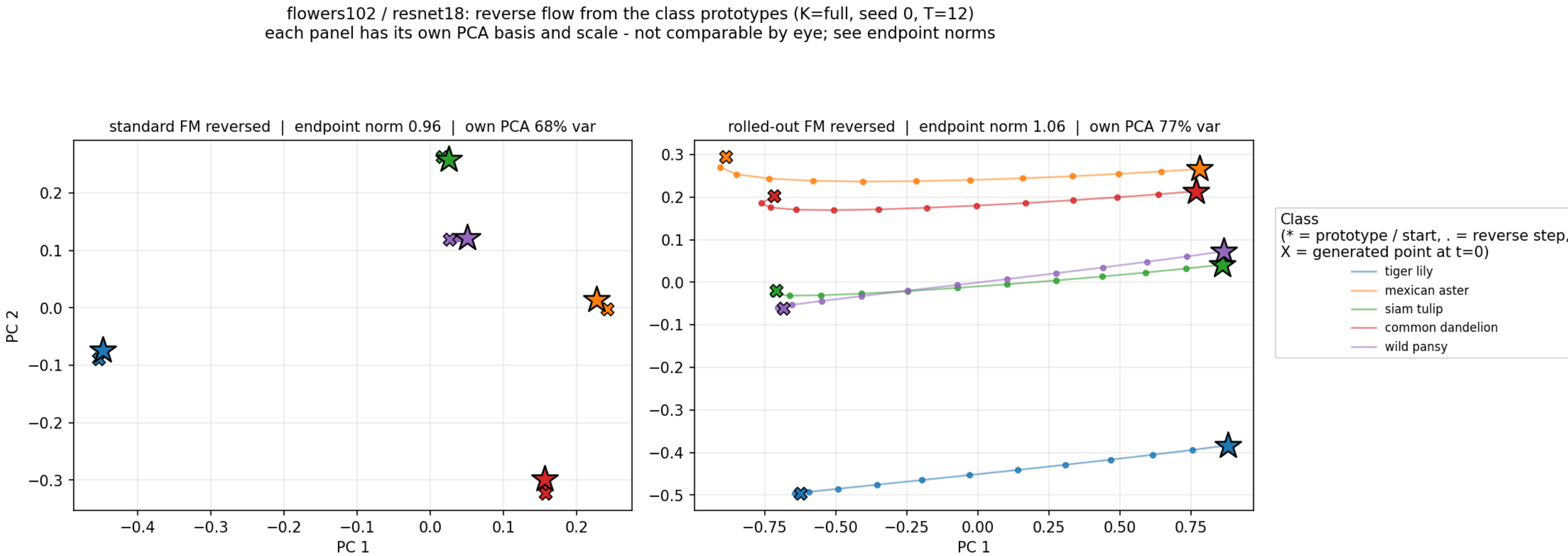
# Stage 2: reverse flow from the class prototypes (PCA)

## *dtd / resnet18*



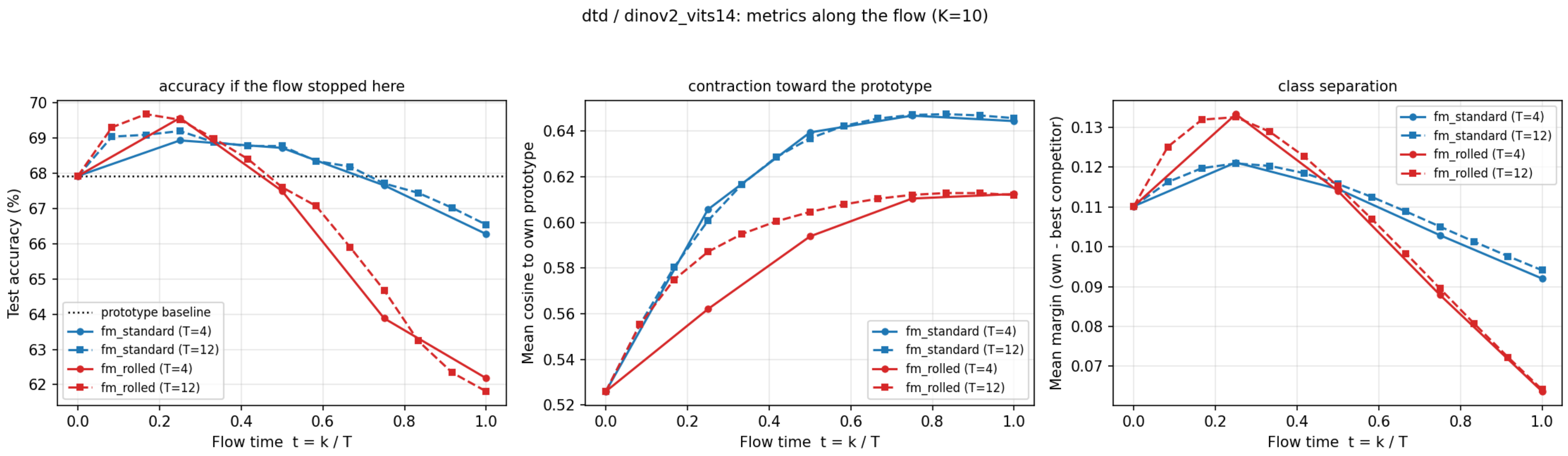
## Stage 2: reverse flow from the class prototypes (PCA)

### *flowers102 / resnet18*



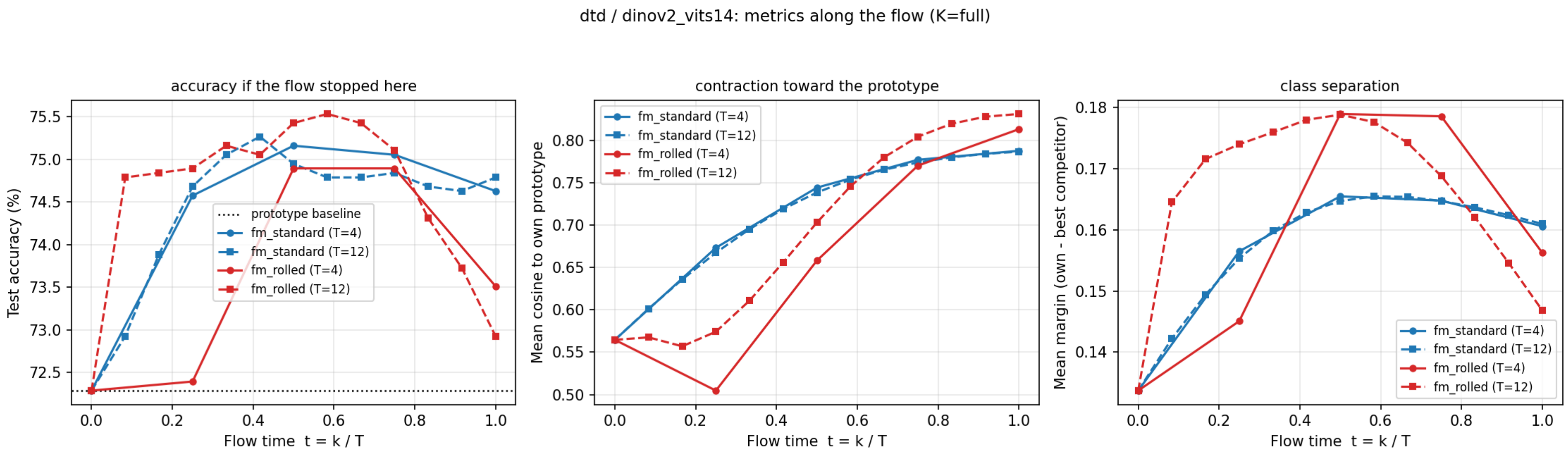
# Stage 2: metrics along the flow

## *dtd / dinov2_vits14 (K=10)*



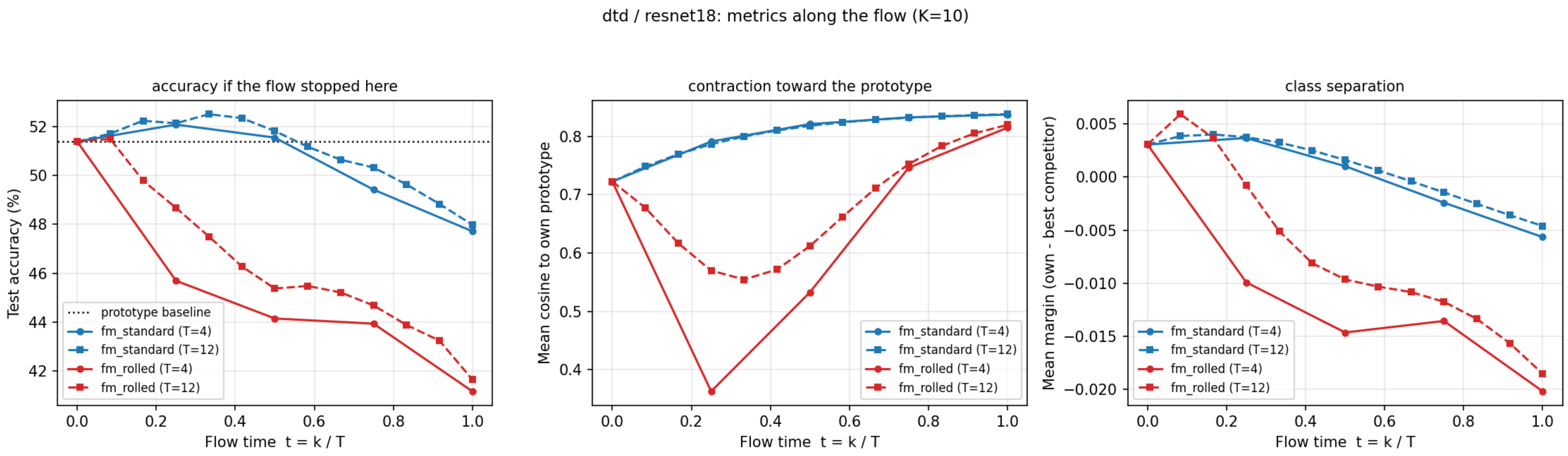
# Stage 2: metrics along the flow

## *dtd / dinov2_vits14 (K=full)*



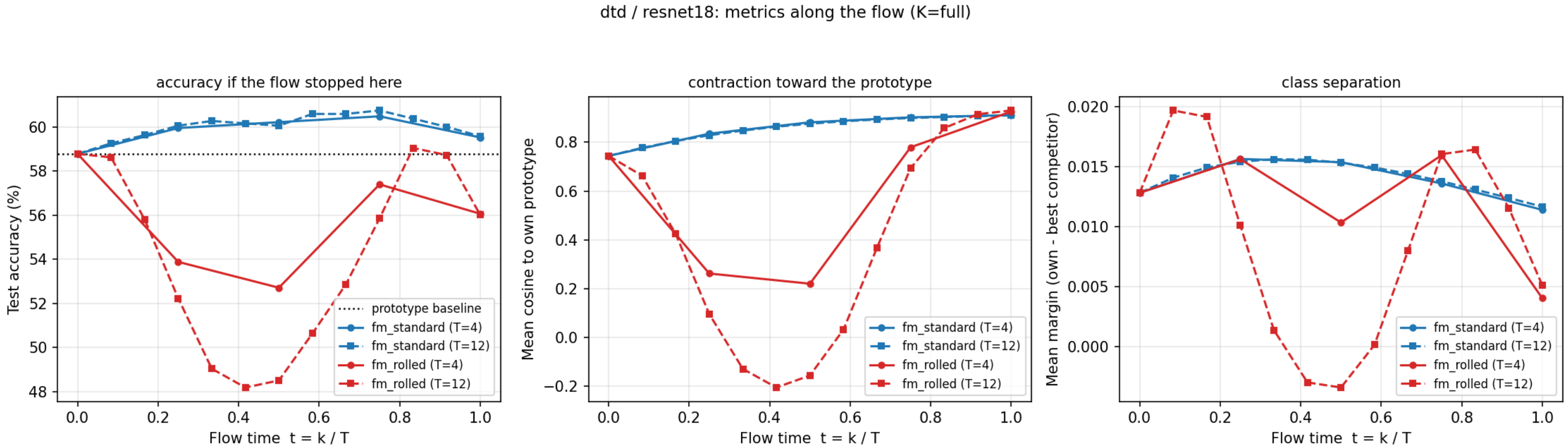
## Stage 2: metrics along the flow

### *dtd / resnet18 (K=10)*



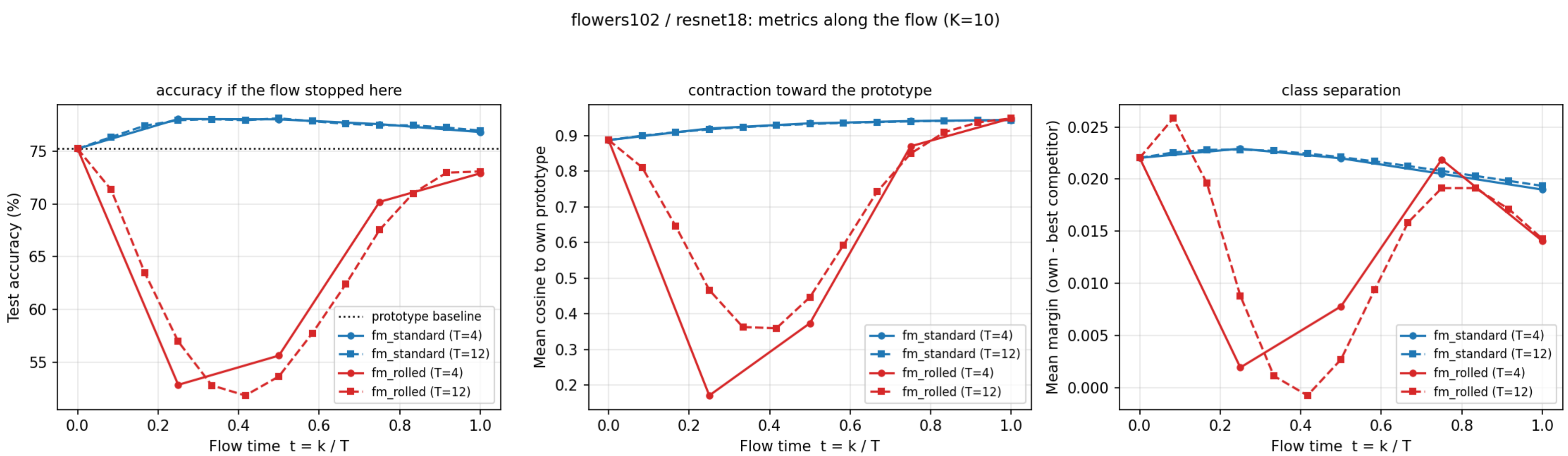
# Stage 2: metrics along the flow

## *dtd / resnet18 (K=full)*



## Stage 2: metrics along the flow

### *flowers102 / resnet18 (K=10)*



# Stage 2: metrics along the flow

## *flowers102 / resnet18 (K=full)*